Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **C++/Python-ready systems**, imaging-adjacent ML pipelines, hardware-integration tooling, technical documentation, Agile delivery, and reliable software interfaces.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform integration services** with OOP patterns, Python, SQL, AWS Glue, and S3, improving response time by **30%** across validation, release, operational handoffs, hardware-style interfaces, and reports.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across integration flows, increasing CI reliability by **68%** while reducing deployment regressions, defects, rework, and support risks.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python services and REST APIs** handling 10,000+ daily requests, implementing file ingestion, concurrent processing, automated checks, documented fixes, sprint delivery, triage, release reviews, and support.
- Partnered with users to debug **workflow and execution defects** by reproducing OCR, persistence, and processing failures, documenting regression cases, reviewing fixes, and cutting feedback turnaround by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **ML evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and visual summaries for reproducible experiments, model-analysis workflows, diagnostics, reporting, image-style validation, reviews, and dashboards.
- Documented **prompt, evaluation, and reporting workflows** for researchers, turning ad hoc experiments into repeatable processes with clearer technical handoffs, validation outputs, analysis notes, summaries, reports, and reviews.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **modular operational services** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, integrations, monitoring, debugging, validation, releases, technical documentation, reviews, and reports.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, shared-environment handoffs, verification checks, release coordination, and support.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **database-backed software tools** for 500+ users, translating requirements into APIs, normalized schemas, monitored jobs, technical documentation, validation logic, and production releases across cloud storage systems reliably.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting production handoffs and reviews.

Skills

Languages: C++, Python, C#/.NET familiarity, Java, SQL, JavaScript, TypeScript, Bash, HTML/CSS
ML/Imaging: PyTorch/TensorFlow familiarity, computer vision exposure, image/data pipelines, model evaluation
Systems: REST APIs, WebSockets, hardware interface readiness, motion/robotics interest, debugging, scheduling
Tools: Visual Studio familiarity, GitHub, coding agents, Docker, Jenkins, CI/CD, Agile, issue tracking
Delivery: Technical design docs, training documentation, reports, validation, customer communication, release support

Projects

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*

- Benchmarked **BLIP, GIT, ViT-GPT-2, and custom vision models** on 10K images and 100K captions, improving caption accuracy from **81% to 96%** with reproducible evaluation reports and metrics.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript, OS Algorithms*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, React controls, Gantt visualization, dynamic timing calculations, debugging workflows, validation, simulations, analysis, metrics, and tests.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized Python REST API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting and automated tests reliably.